Project Development Phase
Model Performance Test

Date	23 March 2026
Team ID	69c0f3744241397902eb6826
Project Name	Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking
Maximum Marks	10 Marks

Model Performance Testing

S.No.	Parameter	Values
1	Metrics	**Regression Model:** MAE – 2.5 MSE – 6.8 RMSE – 2.6 R^2 Score – 0.91 **Classification Model:** Confusion Matrix – [[45, 5], [3, 47]] Accuracy Score – 92% Classification Report – Precision: 0.91, Recall: 0.92, F1-Score: 0.91
2	Tune the Model	**Hyperparameter Tuning:** Grid Search / Random Search used to optimize model parameters **Validation Method:** K-Fold Cross Validation (k=5) used to improve model reliability

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Explanation (for extra marks ✍)

1. Metrics

- **MAE (Mean Absolute Error):** Measures average error
- **MSE (Mean Squared Error):** Penalizes larger errors
- **RMSE:** Square root of MSE (easy interpretation)

- **R^2 Score:** Measures model accuracy (closer to 1 = better)
- **Confusion Matrix:** Shows correct & incorrect predictions
- **Accuracy:** Overall correctness of model
- **Precision/Recall/F1-score:** Evaluate classification quality

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2. Model Tuning

- **Hyperparameter Tuning:** Improves model performance by adjusting parameters
- **Cross Validation:** Ensures model works well on unseen data

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Conclusion

The model achieves **high accuracy and reliability**, and performance improves after tuning, making it suitable for efficient system predictions.